

Top 50 Linux Commands Reference Guide

Essential command-line tools for system administration, file management, networking, and troubleshooting.

Pro Tip: Most Linux commands support the `--help` flag or can be explored in detail using the `man <command>` visual manual system.

1. FILE & DIRECTORY NAVIGATION

Commands 01 – 10

#	COMMAND	DESCRIPTION & UTILITY	PRACTICAL EXAMPLE
01	ls	List directory contents. Displays files and subdirectories with various details using flags.	<code>ls -la /var/log</code>
02	cd	Change current working directory. Navigates to target path or home directory.	<code>cd /etc/nginx</code>
03	pwd	Print Working Directory. Outputs full absolute path of the current directory location.	<code>pwd</code>
04	mkdir	Make directory. Creates new folders; <code>-p</code> creates missing parent directories as needed.	<code>mkdir -p project/src/bin</code>
05	rmdir	Remove empty directory. Deletes directories that contain no files or subfolders.	<code>rmdir old_empty_folder</code>
06	cp	Copy files and directories. The <code>-r</code> flag recursively copies folder trees.	<code>cp -r /app/config /app/backup</code>
07	mv	Move or rename files and directories from source location to target path.	<code>mv draft.txt published.txt</code>
08	rm	Remove files or directories. Use <code>-rf</code> with extreme care for recursive force removal.	<code>rm -i temp_data.csv</code>
09	touch	Create empty file or update access/modification timestamps of existing files.	<code>touch index.js</code>
10	find	Search files in directory hierarchy based on criteria like name, size, or modification date.	<code>find /var/www -name "*.html"</code>

2. TEXT PROCESSING & FILE VIEWING

Commands 11 – 20

#	COMMAND	DESCRIPTION & UTILITY	PRACTICAL EXAMPLE
11	cat	Concatenate and display contents of files directly to stdout stream.	<code>cat /etc/resolv.conf</code>
12	less	View file contents interactively with backward and forward keyboard navigation.	<code>less /var/log/syslog</code>
13	head	Output the first part (default 10 lines) of files to standard output.	<code>head -n 20 /etc/passwd</code>
14	tail	Output the last part of files. -f monitors file growth continuously in real time.	<code>tail -f /var/log/nginx/error.log</code>
15	grep	Search text patterns using regular expressions in files or piped command outputs.	<code>grep -rn "ERROR" /var/log/</code>
16	sed	Stream editor for parsing and transforming text (e.g., search and replace inline).	<code>sed 's/localhost/127.0.0.1/g' app.conf</code>
17	awk	Pattern scanning and processing language for structured columnar data manipulation.	<code>awk '{print \$1, \$3}' data.txt</code>
18	cut	Remove sections from each line of files based on explicit byte or field delimiters.	<code>cut -d':' -f1 /etc/passwd</code>
19	sort	Sort lines of text files alphanumerically or numerically with options like -r (reverse).	<code>sort -n scores.txt</code>
20	uniq	Report or filter out repeated adjacent lines. Frequently combined with sort.	<code>sort list.txt uniq -c</code>

3. PERMISSIONS & OWNERSHIP

Commands 21 – 23

#	COMMAND	DESCRIPTION & UTILITY	PRACTICAL EXAMPLE
21	chmod	Change access permissions for files or directories (read, write, execute modes).	<code>chmod 755 deploy.sh</code>
22	chown	Change file owner and group attributes for specific assets or recursive trees.	<code>chown -R www-data:www-data /var/www</code>
23	umask	Set or view default permission masks applied automatically to newly created files.	<code>umask 022</code>

4. SYSTEM PERFORMANCE & PROCESS CONTROL

Commands 24 – 32

#	COMMAND	DESCRIPTION & UTILITY	PRACTICAL EXAMPLE
24	ps	Display snapshot of current running processes with PIDs, memory usage, and TTYS.	<code>ps aux grep python</code>
25	top	Display dynamic real-time view of running system processes, CPU utilization, and RAM.	<code>top</code>
26	htop	Interactive process viewer (enhanced version of top) with color-coded graphs and scroll.	<code>htop</code>
27	kill	Send termination or custom signal (e.g. SIGKILL -9) to target process via process ID.	<code>kill -9 1234</code>
28	pkill	Send signals to processes matching specified process name patterns instead of PID numbers.	<code>pkill -f nginx</code>
29	systemctl	Control systemd init system and manage background services (start, stop, status).	<code>systemctl status docker</code>
30	journalctl	Query and view log records generated by systemd journal service daemon.	<code>journalctl -u nginx -n 50</code>
31	free	Display amount of free and used memory (RAM and Swap) in human-readable format.	<code>free -h</code>
32	uptime	Tell how long system has been running alongside current load averages (1, 5, 15 min).	<code>uptime</code>

5. DISK STORAGE & FILE SYSTEM MANAGEMENT

Commands 33 – 37

#	COMMAND	DESCRIPTION & UTILITY	PRACTICAL EXAMPLE
33	df	Display total disk space usage across mounted file system partitions in readable units.	<code>df -h</code>
34	du	Estimate disk space usage of files and subdirectories recursively.	<code>du -sh /var/log/*</code>
35	lsblk	List block storage devices (hard drives, SSDs, partitions) in tree layout.	<code>lsblk</code>
36	fdisk	Manipulate disk partition tables (create, resize, delete partitions). Requires root.	<code>sudo fdisk -l</code>
37	mount	Attach storage device or file system partition to specified directory mount point.	<code>sudo mount /dev/sdb1 /mnt/usb</code>

6. NETWORKING & COMMUNICATION

Commands 38 – 44

#	COMMAND	DESCRIPTION & UTILITY	PRACTICAL EXAMPLE
38	ping	Send ICMP ECHO_REQUEST packets to network hosts to test reachability and latency.	<code>ping -c 4 google.com</code>
39	ip	Modern utility to show/manipulate routing, network devices, interfaces, and IP addresses.	<code>ip addr show eth0</code>
40	netstat / ss	Display active network sockets, listening ports, and route tables (ss is modern default).	<code>ss -tulpn</code>
41	curl	Transfer data to or from network servers supporting HTTP, HTTPS, FTP, and REST APIs.	<code>curl -I https://example.com</code>
42	wget	Non-interactive network downloader capable of HTTP/HTTPS/FTP background transfers.	<code>wget https://example.com/file.zip</code>
43	ssh	Secure Shell terminal remote login client for encrypted communication between hosts.	<code>ssh user@remote-server.com</code>
44	scp	Secure Copy protocol to copy files securely over encrypted SSH connections between hosts.	<code>scp file.txt user@remote:/tmp/</code>

7. ARCHIVES, PACKAGE MANAGEMENT & SYSTEM INFO

Commands 45 – 50

#	COMMAND	DESCRIPTION & UTILITY	PRACTICAL EXAMPLE
45	tar	Tape Archiver tool to bundle multiple files into tarballs or extract compressed archives.	<code>tar -czvf archive.tar.gz /data</code>
46	zip / unzip	Compress or unpack standard PKZIP compatible compressed file archives.	<code>unzip archive.zip -d /target</code>
47	apt / yum	Package management utilities (Debian/Ubuntu vs. RHEL/CentOS) to install software updates.	<code>sudo apt update && sudo apt install git</code>
48	uname	Print system architecture, Linux kernel release version, and operating system info.	<code>uname -a</code>
49	whoami	Display current effective user ID name associated with active shell session.	<code>whoami</code>
50	sudo	Execute commands with administrative superuser (root) privileges securely.	<code>sudo systemctl restart nginx</code>

Essential Keyboard Shortcuts

- Ctrl + C : Cancel current running process
- Ctrl + Z : Send process to background
- Ctrl + R : Reverse search bash command history
- Ctrl + L : Clear current terminal screen
- Tab : Autocomplete paths and command names
- !! : Execute last entered command

